

Nuclear Structure Investigations of Light Nuclei with the R3B Experiment

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R³B Experiment @FAIR

Total Interaction Cross Section $^{12}\text{C} + ^{12}\text{C}$

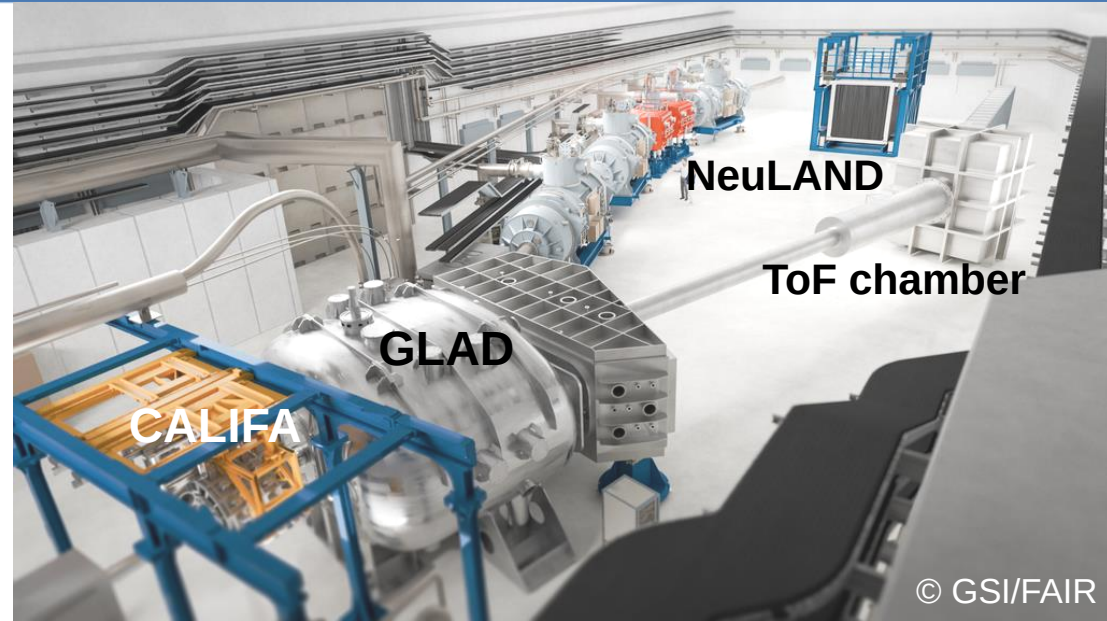
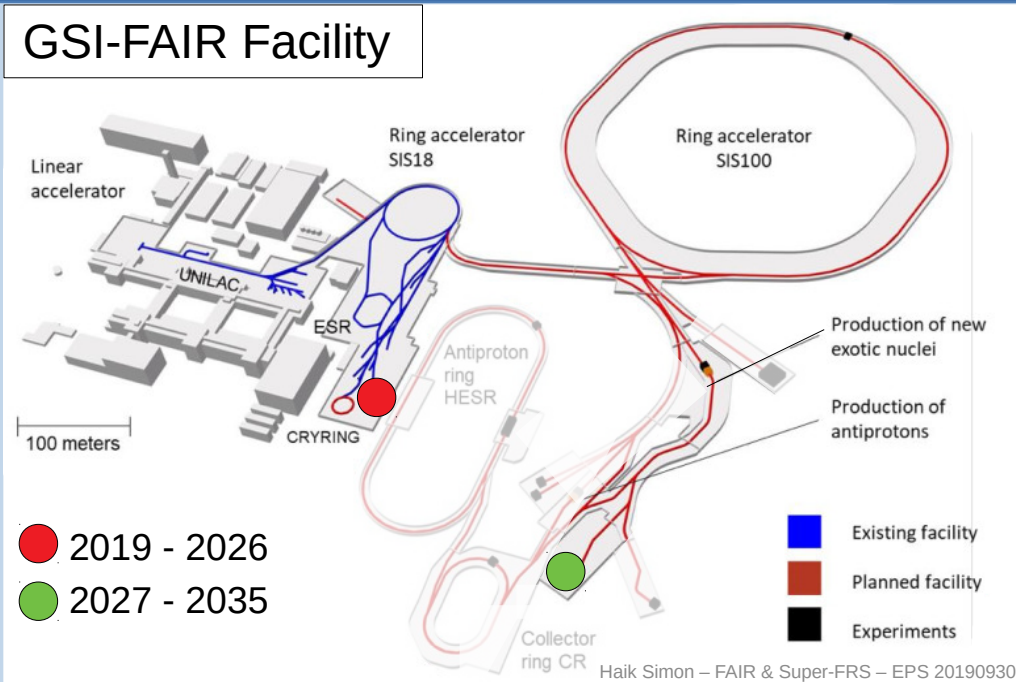
Quasi-Free Scattering $^{12}\text{C}(p,2p)^{11}\text{B}$

Fission Experiments @ R3B

ML for Cluster Reconstruction in CALIFA

Funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation)
under Germany's Excellence Strategy – EXC-2094 – 390783311,
BMBF 05P19WOFN1, 05P21WOFN1 / BMBF 05P24WO2
and the FAIR Phase-0 program

GSI-FAIR Facility

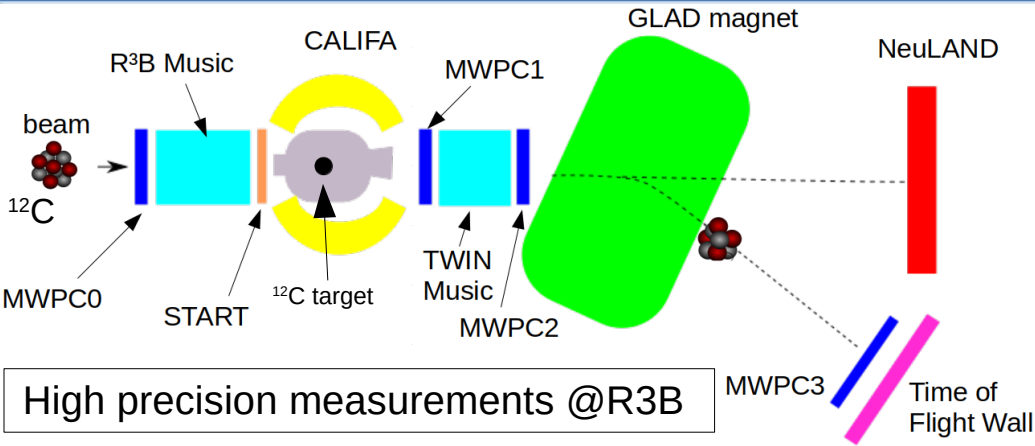


Experimental Approach:

- Inverse kinematics
- Kinematically complete measurements

Research Areas of My PhD Thesis

- Benchmark measurements for the EoS experiments
- Validation of QFS-reaction $^{12}\text{C}(p,2p)^{11}\text{B}$ and complementary fission studies
- Instrumental Development



Motivation:

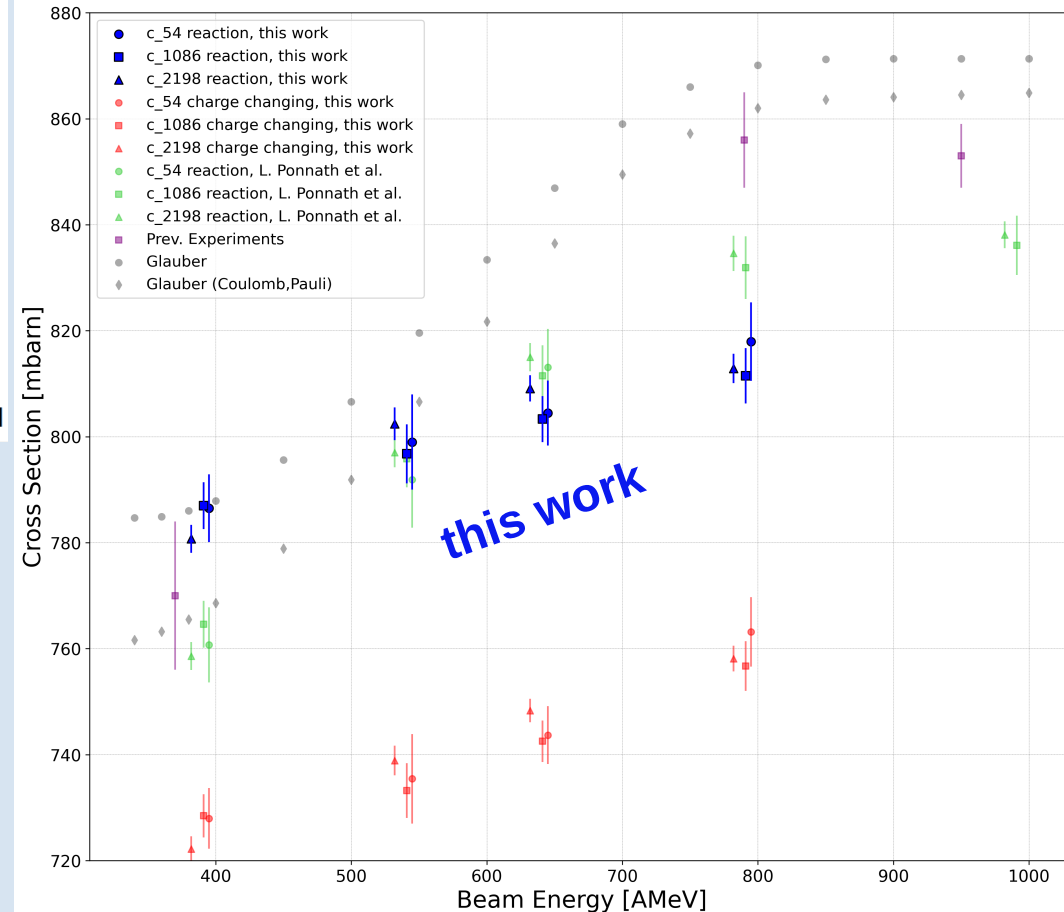
Quantify the uncertainty of the reaction model

Future physics reach:

Constrain slope parameter L of the symmetry energy of the EoS via precise $\sigma_{\Delta N}$ measurements

Measurement via transmission method:

$$N_2 = N_1 e^{-N_t \sigma}$$

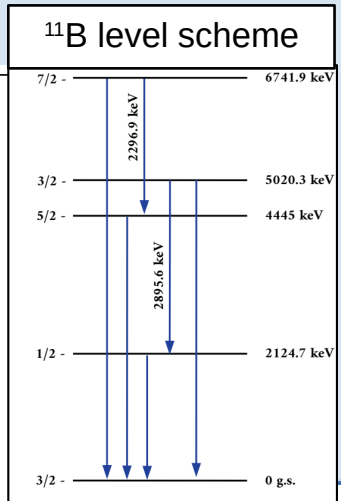
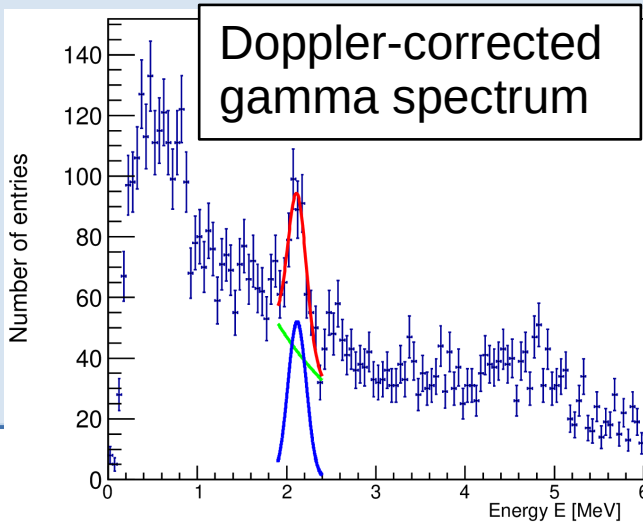
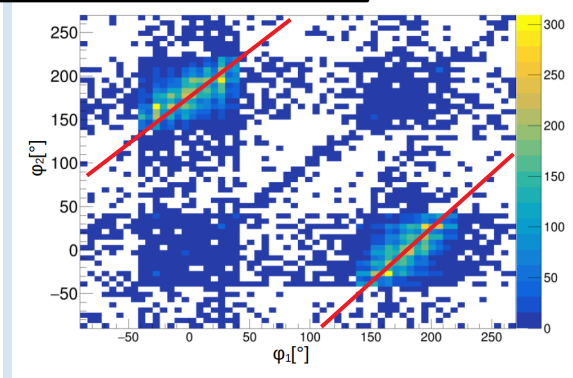
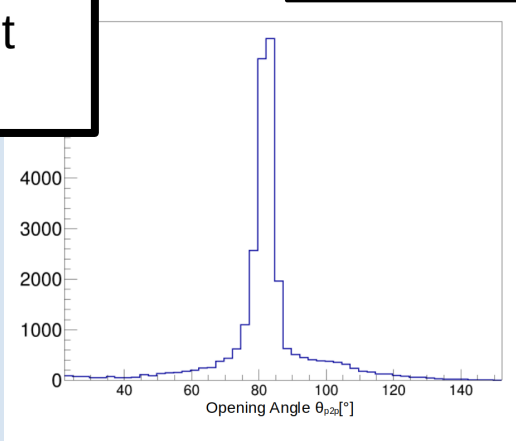
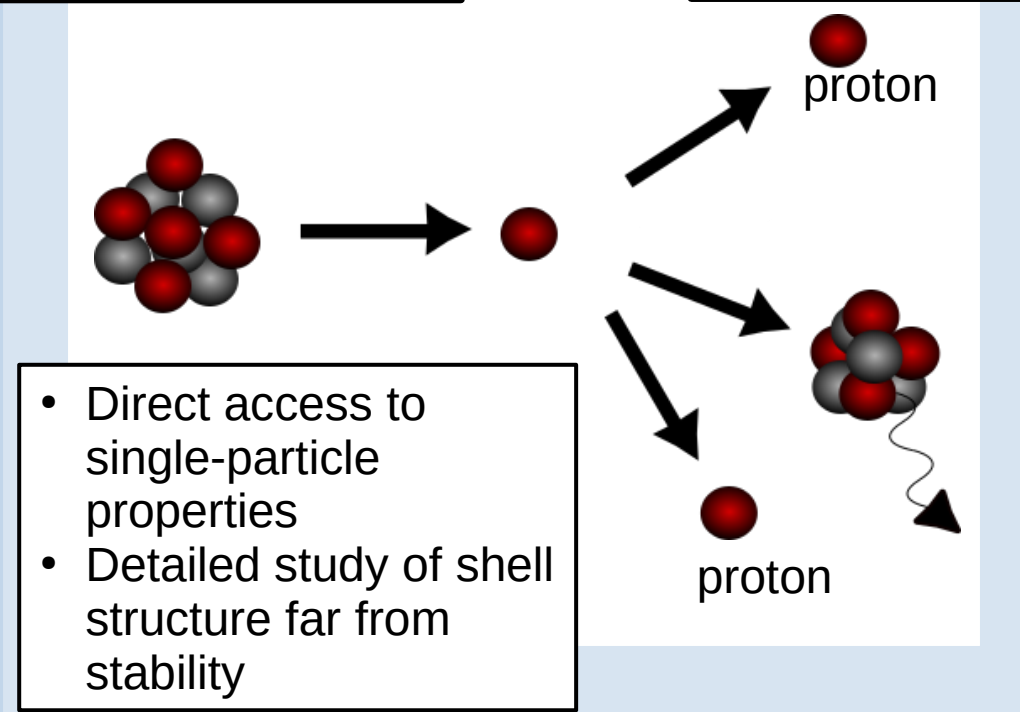


Quasi-Free Scattering $^{12}\text{C}(p,2p)^{11}\text{B}$

➤ ^{12}C beam
➤ proton like target

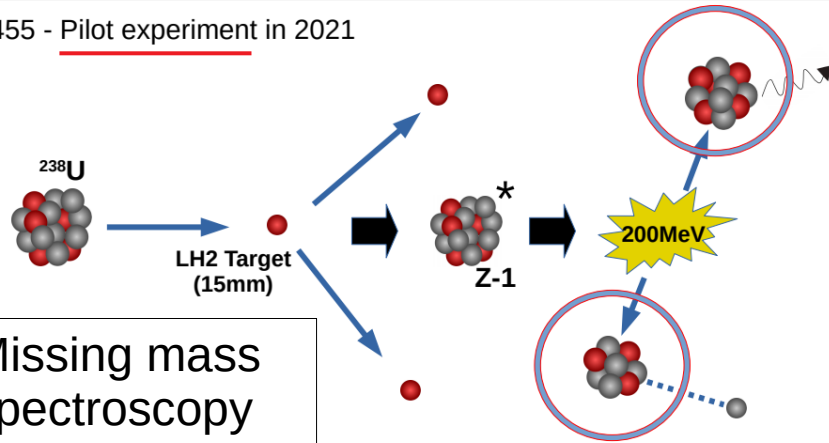
➤ 2 protons
➤ ^{11}B fragment (spectator)

two-proton correlations



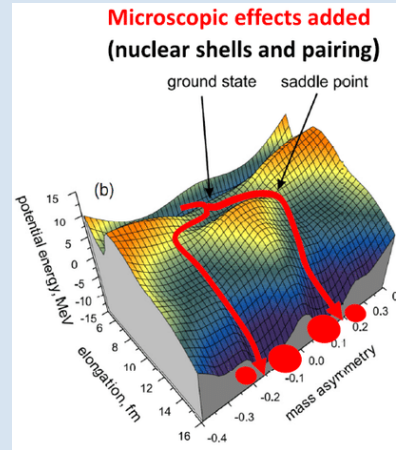
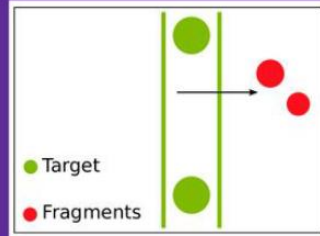
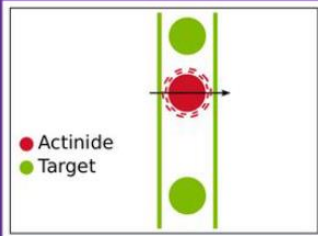
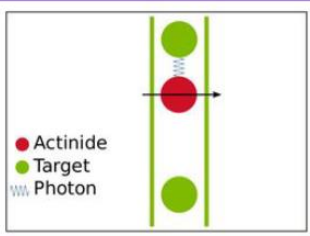
Fission via QFS: $^{238}\text{U}(p, 2pf)$

S455 - Pilot experiment in 2021

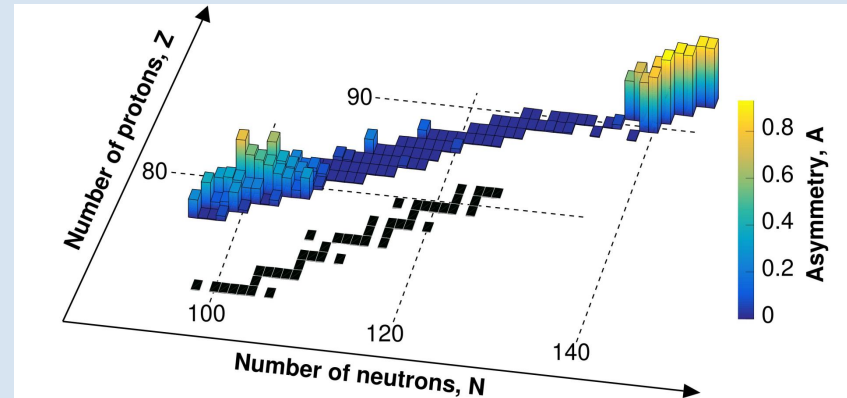
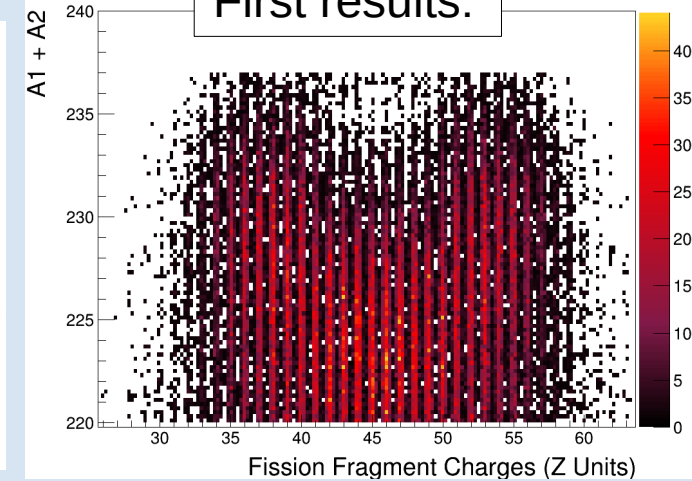


Missing mass spectroscopy

Complementary method via coulomb excitation

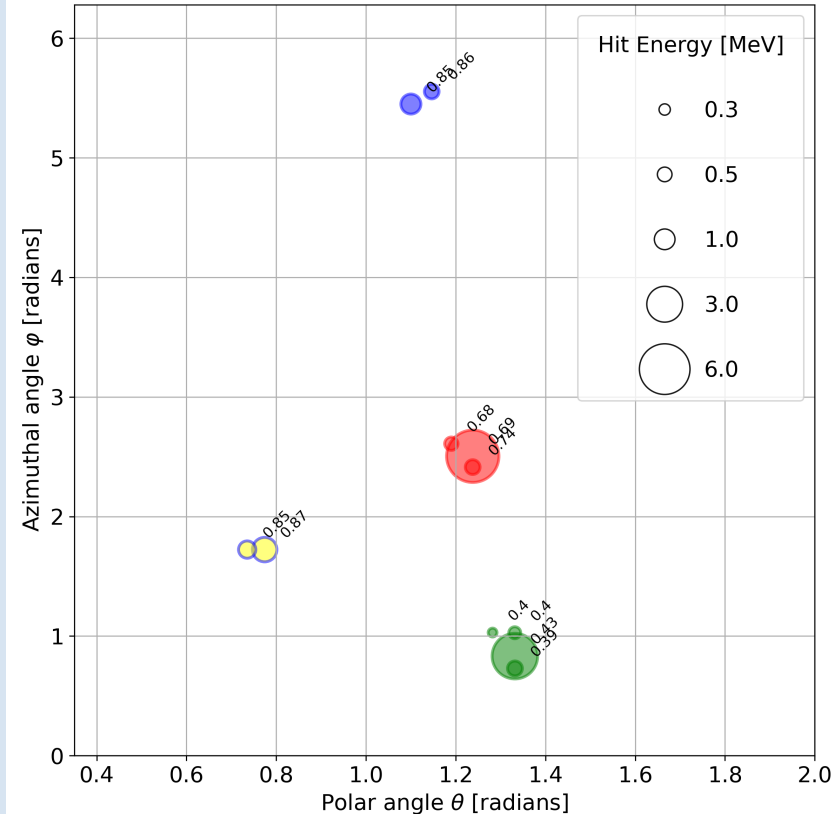
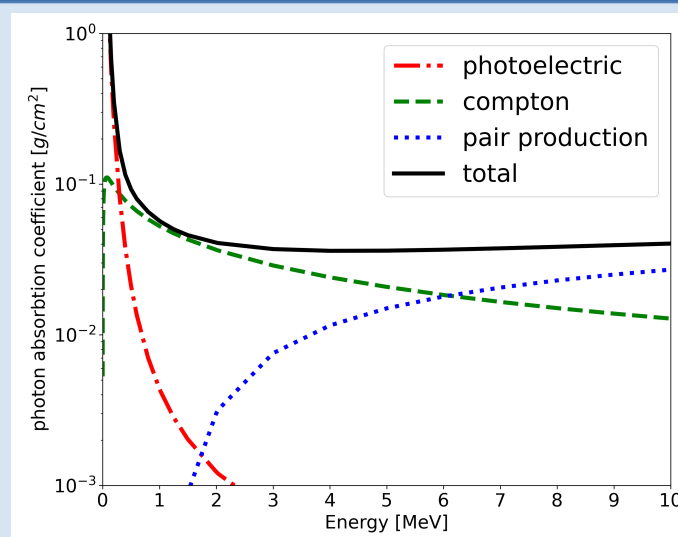
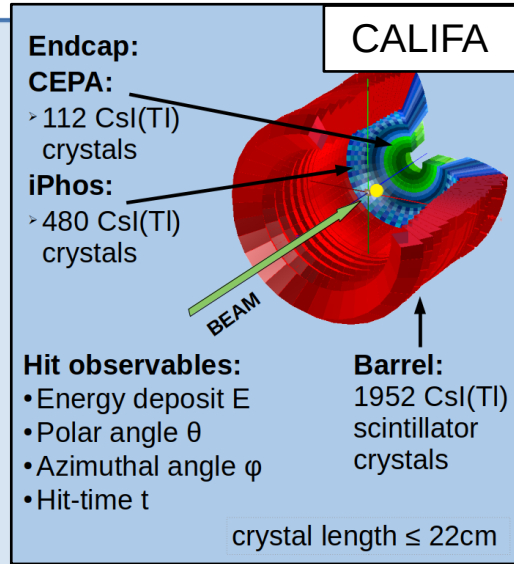


First results:



"An asymmetric fission island driven by shell effects in light fragments", Nature (2025)

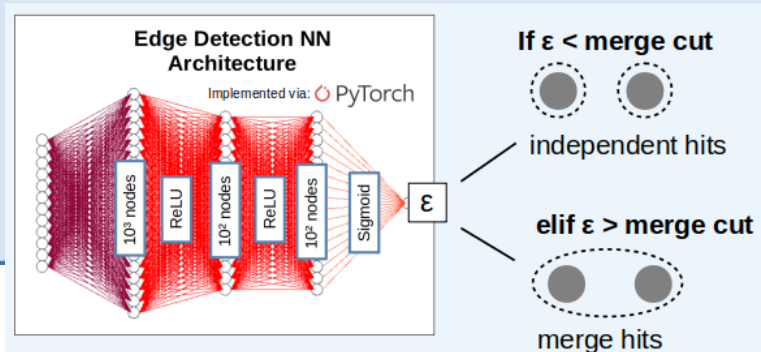
Machine Learning for the Cluster Reconstruction in the CALIFA Calorimeter at R3B



Edge Detection Neural Network

2. Pairwise hit comparison (i, j)

1. Pre- clustering



With support from Origins
Data Science Lab